How much ordering information does a deep mutational scan contain?

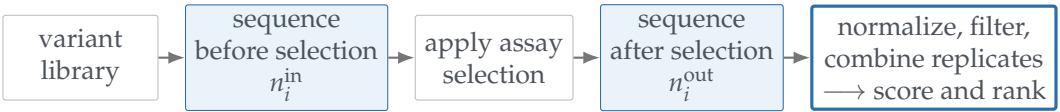
Quantifying ranking resolution in protein fitness landscapes

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Main question: which parts of a DMS ordering are actually supported by the observed counts?

A DMS measures counts; the ranking is constructed



Direct observations

- counts at finite sequencing depth
- replicate, timepoint, or gate structure

Analysis choices

- score definition and normalization
- filtering and replicate aggregation

A published score table is an **analysis output**, not a direct measurement of an assay-independent fitness.

The same point estimate can have very different support

	input	output	ratio
sparse	1	10	10
deep	1,000	10,000	10

A one-read change alters the first ratio substantially, but hardly moves the second.

Two distinct sources of ambiguity

1. **Definition dependence:** reasonable score constructions can produce different orderings.
2. **Finite resolution:** after fixing one analysis, finite counts leave uncertainty about the induced ordering.

> *How much of the ordering induced by a DMS landscape is supported by the observed counts?*

Why ranking resolution matters

Protein engineering

- Which variants should be validated?
- Which mutations should be combined?
- How many candidates are genuinely distinguishable near the cutoff?

Machine-learning evaluation

- Spearman and top-K metrics depend on label ordering.
- A small model difference may be real but unresolved by the experiment.
- Benchmark labels should not imply more precision than the counts contain.

The useful output is often a **candidate set with inclusion probabilities** and a resolution scale, not an exact ranking.

What we quantify, and why the construction is useful

HMC posterior

- Fit a generative count model.
- Sample latent assay-score landscapes.
- Include fitted-score parameter uncertainty.

MLE predictive

- Fix parameters at the MLE.
- Sample repeated count experiments.
- Recompute the source count-derived score.

sampled score landscapes → rank every sample → rank intervals, $q_i(K)$, top-K overlap

The procedures answer **different uncertainty questions**; their numerical intervals are not interchangeable.

Experimental systems and current analysis scope

system	assay/readout	landscapes	fitted variants	HMC runs	MLE-pred. runs
PDZ	peptide binding	5	10.7k–151.6k	14	19
TrpB	enzyme activity coupled to growth	10	9.0k–194.4k	28[a]	34
NucB	fluorescence-sorted nuclease activity	4	10.8k–18.8k	7	7
total		**19**		**49**	**60**

[a] HMC covers nine TrpB landscapes; four-site is MLE-only. Cross-procedure figures use the 18 shared landscapes. Sources: Zarin et al.; Johnston et al.; Thomas et al.

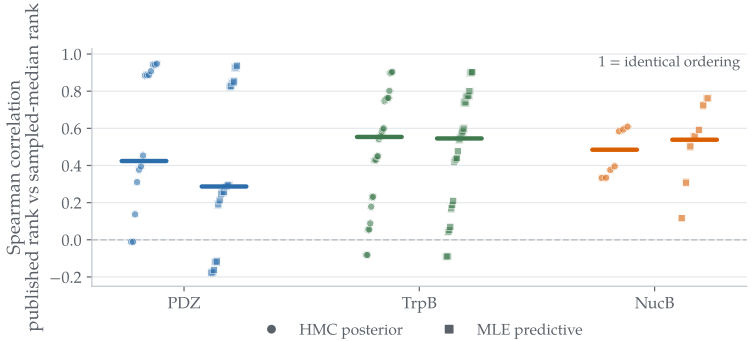
Alignment Raw counts, fitted axes, and candidate/control roles are checked.

Score checks TrpB and NucB source scores are reproduced exactly.

Predictive checks All 60 MLE artifacts pass shape and fixed-total checks.

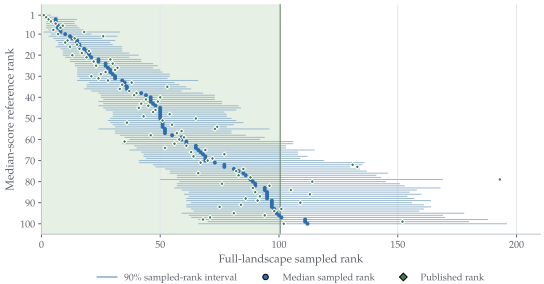
PDZ MLE results use a provisional DiMSum-style proxy.

The published ordering is not always recovered



Points are analysis runs; bars are landscape-weighted medians. Agreement ranges from near zero or slightly negative to above 0.9.

A precise top ten is not ten precisely resolved positions

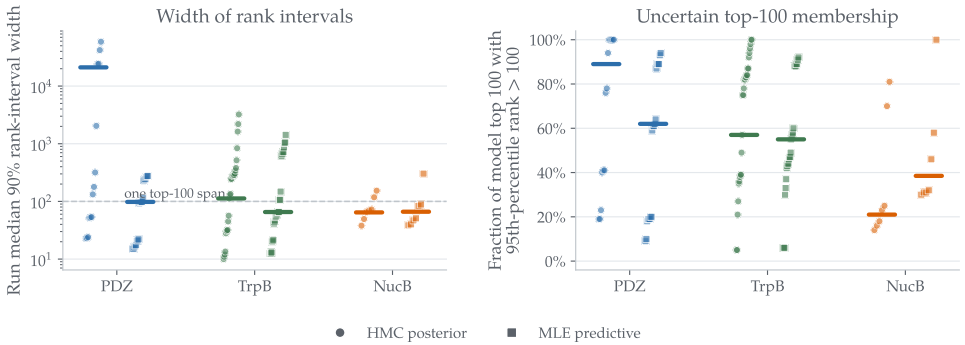


PDZ cis example

- Rows are ordered by median fitted score.
- Lines show 90% sampled-rank intervals.
- Several top-ten intervals overlap and extend beyond rank 10.

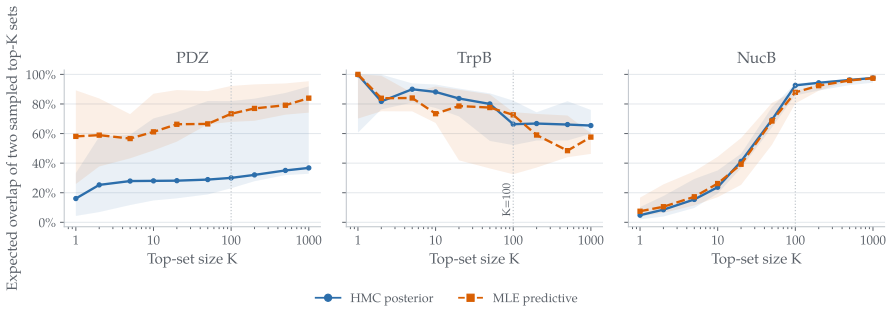
The experiment can identify a promising region without resolving the exact order inside it.

Top-100 rank resolution varies by system and analysis



Landscape-weighted median top-100 exit fractions: HMC **89% PDZ, 57% TrpB, 21% NucB**; MLE predictive **62%, 55%, 39%**.

Decision stability depends on both system and set size

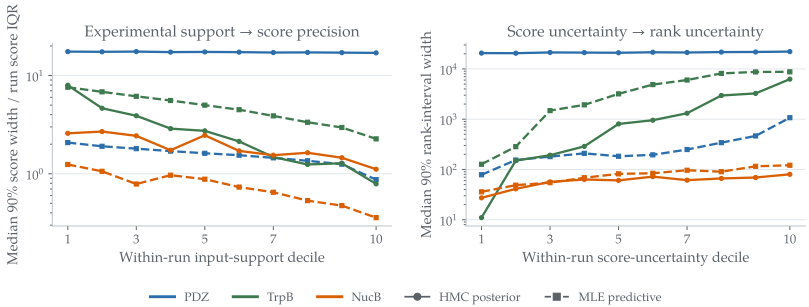


Metric: expected fraction shared by two sampled top-*K* sets.

Reading: *K* is part of the scientific decision, not a plotting choice.

Resolution can change sharply with *K* and differs across uncertainty procedures.

What is associated with ranking resolution?

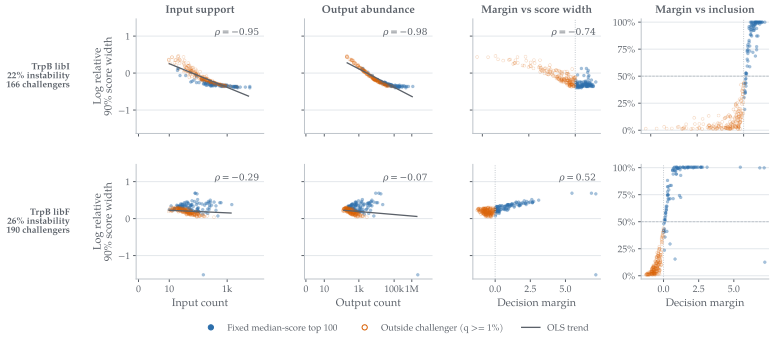


Higher input support is generally associated with narrower score intervals.

Wider score intervals and local crowding are associated with wider rank intervals.

Descriptive associations only; they do not estimate the causal effect of adding reads.

Similar instability can arise from different local patterns



Same protein, model, and uncertainty procedure; similar top-100 instability. LibI shows strong support–precision coupling and $\rho_{\text{margin,width}} = -0.74$; libF shows weak count coupling and $\rho_{\text{margin,width}} = +0.52$.

Deliberately selected Figure 10 contrast. Primary cohort = fixed top 100 plus outside candidates with $q_i(100) \geq 0.01$. Decision-margin ranges are analysis-specific; exact zero widths use a plot-only floor.

What we can conclude, and what remains conditional

Supported by the current analyses

- DMS orderings have finite, heterogeneous resolution.
- Reported and fitted central orderings can differ.
- Top-K uncertainty can be expressed directly as decision probabilities.
- Count support and score geometry help explain where resolution is lost.

Not implied by the current analyses

- a universal, assay-independent true fitness
- MLE-predictive intervals as MLE confidence intervals
- a causal estimate of how much more sequencing would help
- that a published ranking is simply "wrong"

Recommendation: report rank intervals, top-K inclusion probabilities, and the assay/analysis definition with point scores.

Discussion: which decision should define K, and how should label resolution enter benchmark comparisons?

Backup: the three decision-level estimands

For sampled top-K set $T_K^{(s)}$, fixed median-score set $T_K^\star$, and published set T_K^{rep}:

$$C_K = E_{s \neq s'}\left[\frac{|T_K^{(s)} \cap T_K^{(s')}|}{K}\right] \qquad \text{sample-to-sample reproducibility,}$$

$$C_K^\star = E_s\left[\frac{|T_K^{(s)} \cap T_K^\star|}{K}\right] \qquad \text{agreement with the fixed analysis reference,}$$

$$O_K^{\mathrm{rep}} = E_s\left[\frac{|T_K^{(s)} \cap T_K^{\mathrm{rep}}|}{K}\right] \qquad \text{agreement with the published top } K.$$

C_K and $C_K^\star$ measure internal decision stability. O_K^{rep} combines internal uncertainty with published-versus-model disagreement.

Backup: how the two uncertainty procedures differ

	HMC posterior	MLE predictive
sampled object	latent assay-score landscape	repeated counts, then source score
parameters	sampled from posterior	fixed at optimized values
target	uncertainty under model and data	repeatability at fixed parameters
not included	assay-independent truth	MLE estimator uncertainty
current scope	49 runs, 18 landscapes	60 runs, 19 landscapes

Similar conclusions are useful; numerical intervals are not estimates of one common variance.

Backup: validation and current limitations

Checks completed

- 60/60 sampled-MLE count artifacts pass direct HDF5 checks
- all TrpB and NucB count-derived score oracles pass
- candidate/control roles and fitted order are explicit
- all proxy, compact-result, and mechanism bundles validate

Limitations to retain

- PDZ MLE score remains a provisional DiMSum-style proxy
- procedures quantify model-conditional, not universal, resolution
- mechanism associations are descriptive
- controlled read downsampling requires altered counts and refitting